Evaluation for Selecting Method Using Reinforcement Learning with Hand-Coded Rules in RoboCup Soccer Agents

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Abstract. Machine-learning algorithms Reinforcement Learning is one of powerful machine learning algorithms. We have known that this algorithm is the effect of robotic learning and that a lot of researchers have proposed the basic idea using this algorithm. A system in which the algorithm has been built is needs a lot of trials and errors. Moreover it requires a huge amount of calculation in order to achieve some effectiveness. This paper has proposed a method using reinforcement learning with hand-coded rules for a soccer agent in RoboCup Soccer Simulation League. We have analyzed a lot of scenes that agents have made scores in simulation soccer games and have extracted some rules. We have called them "hand coded rules". Moreover we have applied our method for an offensive soccer agent in RoboCup Soccer Simulation League and have done some experiments. From the results, we have confirmed that our team has improved its capability of getting scores.

1 Introduction

Some researchers have reported effectiveness of their system installed machine-learning algorithms[1], [2]. We have already known that reinforcement learning is the effects for a lot of applications. For example learning for robotic actions. However there are some problems in the learning algorithm. One of them is a huge amount of calculation processing. To achieve its goal by their system is to maximize a numerical reward signal and the system needs a lot of trials and errors to do so. Therefore this paper has proposed a method using reinforcement learning with hand coded rules. Moreover we have implemented our method for a soccer agent in RoboCup Soccer Simulation League. This paper has also reported results and considerations of evaluation experiments using the agent that is implemented our method.

RoboCup is an international joint project to promote AI, robotics, and related field. A goal of this research, RoboCup Soccer, is to develop a team of fully autonomous humanoid robots that can win against the human world champion team in soccer by 2050. Soccer is one of the most popular sports in the world. Human soccer players need to select for their actions and need to do the actions in real time on their soccer games. For examples, a dribble, sending a pass, a shoot, a tackle and so on. We think that an ability to select actions can be developed by some factors. One of the factors is

a large amount of experience in soccer games and another is a good imagination for their actions to do the excellent actions and so on.

2 Basic Idea

A goal of our research is to figure out a learning ability of human and to put the ability to practical uses. Our basic idea for learning agents' action is that there are some learning methods and sources. In this paper, Fig. 1 illustrates the basic idea for agents' actions. We have considered that the sources of agents' actions can be classified into three groups. In the figure, (1) and (2) mean these groups of rules and the structure of them.

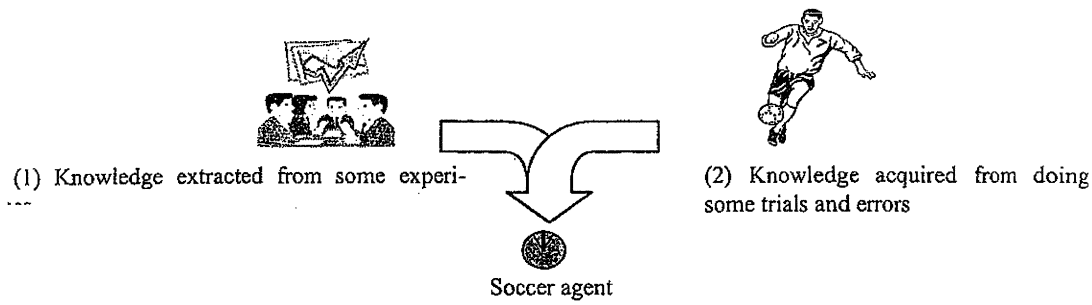


Fig. 1 A Soccer agent has improved its ability by some knowledge from a lot of experiences and some trials and errors

In human soccer games, we consider that action rules in Group (1) are comparable to instructions or strategies from other persons, for example coaches in their team. To realize these cases, we consider that engineers can give soccer agents a part of these rules explicitly. In this research, the rules in Group (1) are hand-coded rules. We have developed the rules from our considerations of results in log files of some RoboCup Soccer simulation games. For the human soccer, we consider that action rules in Group (2) are comparable to empirical rules in soccer players. Human soccer players that have a lot of experiences of soccer games can adapt to some situations in games and take adequate actions. In other cases, players can get improve their skill of soccer through trials and errors during their training. In our research, the action rules in Group (2) are rules that are acquired from examples of actions using unsupervised learning.

In the case that an agent uses the only rule among Group (1), it could do fixed actions and could not have an ability of adapting to dynamic situations. On the other hand, in the case that an agent uses the only rule among Group (2), it could adapt to some cases and would require a lot of trials and errors and a lot of calculation. Therefore, our system has used these two types of rules and has aimed to adjust tradeoff between them using unsupervised learning. Then the agent based on our method has done online-learning and has adapted an opponent team.

3 Our Method

3.1 Hand-Coded Rules

The competition RoboCup Japan Open 2004 was organized in Osaka in May 2004. Log files of games in RoboCup Soccer simulation leagues are opened in web pages [4]. There are 50 games in first round, 30 games in second round and 8 games in final tournament. We have employed some log files of simulation games in second round and in final tournament as sources of rules on the ground that there are wide differences of scoring in some games of first round. Moreover the log files that resulted in nil-all draw are removed from the sources of action rules. The remaining number of log files is 20 files. The total number of scores is 59 in these log files of games. We have discussed contributing factors of their scores in the soccer games. Table 1 has shown a classification of them.

The RoboCup Soccer team YowAI [5] is the champion of RoboCup Soccer Simulation League Japan Open 2004. There are the 7 games, in which the team makes 14 scores, among above 20 log files. We have considered actions of winger just before making the scores. Table 2 has shown a classification of them and figure has indicated Area (1), (2) and (3) that have been used in table 2.

Table 1. Contributing factors of scores

Contributing factors of scores	A number of times (percent)
(a) Crossing pass from a right or left player	26 (44.1%)
(b) Interchanging short pass	21 (35.6%)
(c) Dribble and shoot	5 (8.4%)
(d) Middle or long range shot	0 (0.0%)
(e) Others	7 (11.9%)
Total number	59 (100.0%)

Table 2. A classification of agent actions

Actions	Area (1)	Area (2)	Area (3)	Total
(a) Dribbling	8	8	1	17
(b) Through pass	0	1	0	1
(c) Sending square ball	0	6	6	12
(d) Back pass	0	0	0	0
Total	8	15	7	30

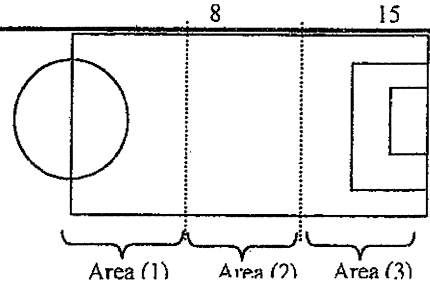


Fig.2 Indicating Area (1), (2) and (3) used in Table 2.

From these considerations, actions for winger have been selected a dribble, a centering that means that a winger sends a ball to a center forward and a back-pass in Area (3). The number of chances for winger's actions in Area (2) is max of them. However actions in Area (3) engage with making scores. Therefore we have limited the actions for left-winger in Area (3) and have applied the agent to our method.

3.2 Selecting Agents' Actions

Fig. 3 has shown a process for selecting a dribble using some numerical values in Area (3). We call one of values threshold in this figure.

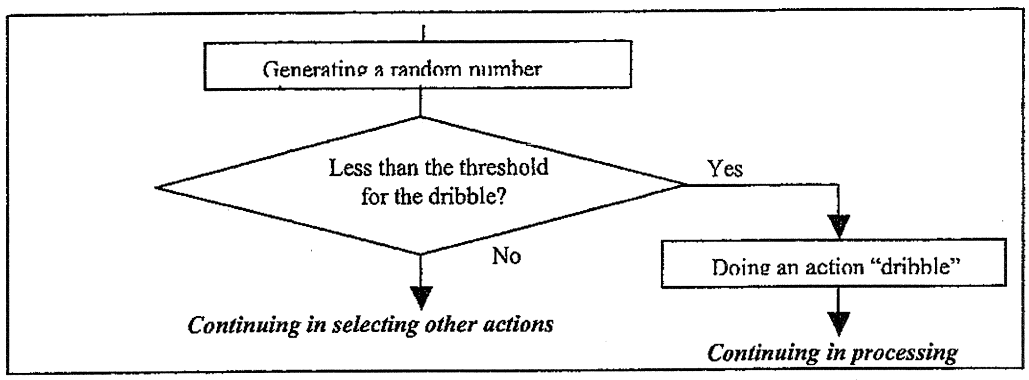


Fig. 3 A flow chart for a process of selecting a dribble.

Then we have employed some values to select an agent's action on our method. Formulas (1), (2) and (3) have indicated these numeric values. In the formulas, a number "i" means the number of each action and they are decided in advance. A value of threshold is summed up likelihood. Then each threshold has been referred from small one to big one. A value of likelihood is an evaluation value over the summation of all of evaluation values. The initial value for calculating the evaluation value in formulas (1) is decided in advance.

Moreover rewards are based on the algorithm of reinforcement learning. This paper has addressed feedback process in order to calculate rewards for agents' actions in sub-section 3.3.

$$EvaluationValue_i = InitialValue_i + \sum \mathrm{Re}ward \quad ...(1)$$

$$Likelihood_i = \frac{EvaluationValue_i}{\sum EvaluationValue} \quad ...(2) \qquad Threshold_i = \sum_{k=1}^{i} Likelihood_k \quad ...(3)$$

3.3 Feedback Process

Our system has given each action's evaluation value each reward in answer to changes of play mode in a simulation game. There is some play mode in the soccer game [6]. Table 3 has shown the rewards that have depend on each play mode.

In the case that winger does one or some actions and an agent in our team can make a score, the system adds 1 into the evaluation value. The reward is max of the

rewards. Our method in this research aims to improve the ability of making scores and the ability of offensive actions. A making a score is the best results. Then we have decided the maximized reward to making a score by the agent in our team. In the case that winger does one or some actions and an agent in our team can get a corner kick, the system adds 0.3 into the evaluation value. In the case that winger does one or some actions and an agent in our team receives an offside, the system adds 0.1 into the evaluation value. In the case that winger does one or some actions and an agent in adversary team can get a goal kick, the system adds 0.1 into the evaluation value. Making scores by our soccer agent is the best result in human and simulation soccer games. However there are scarce numbers of making scores in the soccer games. Then we have decided that the system on our method gave rewards when agents in our team resulted in more effective play modes. From these rewards our system has worked on feedback processes on the algorithm of reinforcement learning.

Table 3. Rewards given by our system in answer to play mode.

Play mode	Rewards
Goal (our team)	1.0
Corner Kick (our team)	0.3
Offside (our team), Goal Kick (adversary team)	0.1
Other modes	0.0

4 Evaluation Experiments

4.1 Procedures

We have done some experiments to confirm our method. For the experiments, we prepared two soccer teams. One of them is ANCT 2004 [7]. This team had a hand in the competition RoboCup Soccer Japan Open 2004 [4]. The team has developed on the team UvA Trilearn [8]. We call the team "Team (A)". Another team is the same as ANCT 2004 and the agent in left-side winger is implemented our method. The winger can make a choice of action from dribble, sending a ball to a center forward and a back-pass. We call the second team "Team (B)". We have done 20 games between Team (A) and (B).

4.2 Results

Team (B) won 12 games, lost 5 games and tied 3 games in these 20 games. Moreover table 4 has shown the total number and the average of making scores. Table 4 has indicated the effectiveness of our method.

Table 4. The total and the average of scoring in 20 games.

	Team (A)	Team (B)
Total	19	42
Average	0.95	2.10

4.3 Considerations

The number of simulation steps in one simulation game is 6000 steps. Table 5 has indicated the averages of making scores in every 1000 steps on each game. From this consideration we have clarified that Team (B) whose winger is implemented our method can improve the ability of making scores in advancing simulation steps. However both teams does not administer stamina in each soccer agent. Therefore their agents cannot keep their stamina and both teams cannot make a lot of scores in last 2000 steps of games. We have to make a study on our method including the administration's problem in the near future.

Table 5. Averages of making scores in every 1000 steps in 20 games

	1000	2000	3000	4000	5000	6000	Total
Team (A)	0.15	0.15	0.15	0.20	0.05	0.25	0.95
Team (B)	0.05	0.55	0.40	0.60	0.30	0.20	2.10

5 Conclusion

This paper addressed our method for reinforcement learning with hand-coded rules. We described our basic idea and hand-coded rules for this method. Then we explained the processes on our method. At last we indicated the experiments and the considerations. From the results we confirmed the effectiveness of our method. Moreover we could figure out the problem of administrations for agents' stamina when the agent has been implemented our method. We need to discuss this problem in the near future.

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